

The Green Premium in German Housing

Econometric Analysis of Energy Efficiency

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Section 1

Introduction

Research Question

How is energy efficiency reflected in advertised rental prices for apartments in Germany?

- Do tenants pay a premium for energy-efficient apartments?
- Does this relationship differ between urban (tight) and rural (loose) rental markets?

Motivation & Context

- **Rising energy costs:**

In 2022, energy prices increased by **34.7%** (Destatis, 2023), making heating costs a key concern for tenants.

- **Market transparency:**

Since **2014**, Energy Performance Certificates (EPCs) are mandatory in housing advertisements (BMW_i, 2014).

- **Related literature:**

- Energy efficiency is capitalized into housing prices in owner-occupied markets (Taruttis & Weber, 2022).
- Evidence for the **rental market**, especially regarding **urban–rural differences**, remains limited.

Section 2

Data

Data Source & Sample

- **Data source:**
ImmoScout24 (RWI Real Estate Data – Campus File, 2022)
- **Unit of observation:**
Individual rental apartment listings
- **Sample construction:**
 - Cross-sectional dataset
 - Exclusion of outliers (rent and size)
 - Removal of observations with missing energy information

Variables

- **Outcome variable:**

Advertised net cold rent per m²
 $\ln(\text{Rent})$

- **Main explanatory variable:**

Energy efficiency class (A+ to H)
Reference category: **Class C**

- **Control variables:**

- Apartment characteristics: size (m²), rooms, building age
- Amenities: balcony, elevator, fitted kitchen, parking, garden

Market Classification

- **Urban markets:**

Single-municipality districts (*kreisfreie Städte*)

- **Rural markets:**

Multi-municipality districts (*Landkreise*)

Descriptive Statistics (Urban vs. Rural)

Table 1: Summary Statistics by Region

Region	Obs	Avg Rent (€/sqm)	Avg Size (sqm)	Efficient (%)
rural	12692	9.51	72.1	31.9
urban	13792	10.05	66.3	27.8

Key Observations

- **Rental prices:**

Urban rents are higher on average (€10.0 vs. €9.5 per m²).

- **Energy efficiency:**

Rural markets exhibit a higher share of energy-efficient apartments (31.9% vs. 27.8%), likely reflecting newer building stock.

Section 3

Methodology

Identification Strategy

Conceptual Framework

- Apartment rents are determined by structural characteristics, amenities, and location.
- Energy efficiency is treated as a **quality attribute** valued by tenants.

Key Assumption

- Conditional on observable characteristics (size, age, amenities, and location), remaining rent differences are associated with energy efficiency.

Estimation

- Ordinary Least Squares (OLS)
- Robust standard errors clustered at the district level

Econometric Model

We estimate the green premium using a hedonic rent regression:

$$\ln(\text{Rent}_i) = \alpha + \beta \text{EnergyClass}_i + \gamma \mathbf{X}_i + \delta_d + \varepsilon_i$$

Model Components

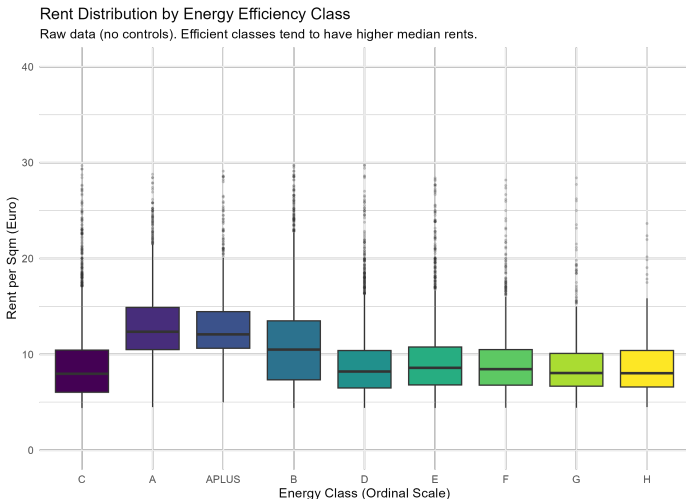
- $\ln(\text{Rent}_i)$: Log rent per square meter
- EnergyClass_i : Energy efficiency class dummies (A+ to H), reference category **C**
- $\mathbf{X}_i$: Apartment characteristics and amenities (age, balcony, elevator, etc.)
- δ_d : District fixed effects (controlling for local price levels)

Section 4

Results

Descriptive Evidence (Raw Data)

Before controls, we see a clear correlation: Better energy classes are associated with higher rents.



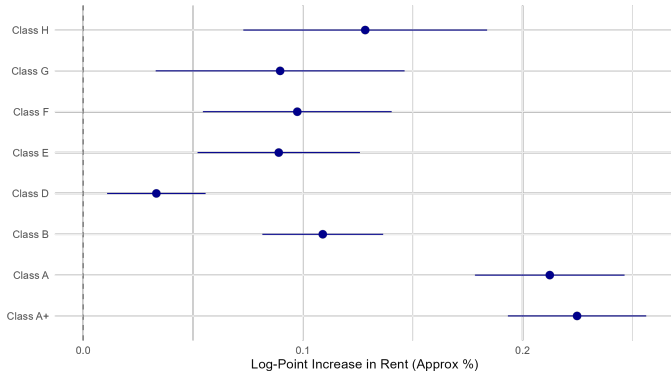
The Green Premium (Regression Results)

After controlling for confounders, the premium remains significant.

- **A+ / A:** Significant premium (Tenants pay ~15-20% more).
- **F / G / H:** Significant discount (Tenants pay less).

The Green Premium: Rent Impact Relative to Class C

Estimates from Hedonic Regression. Bars represent 95% CIs (Clustered SEs).

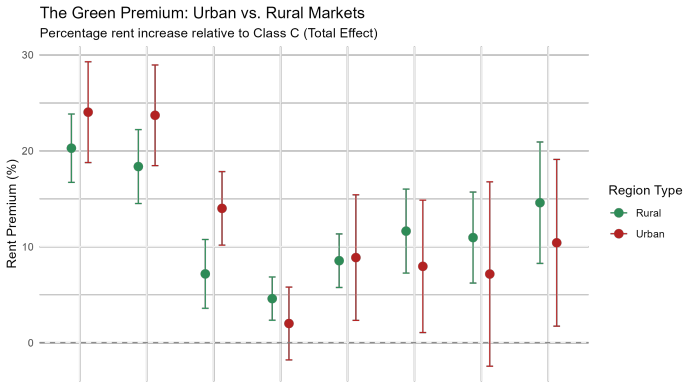


Note: Classes modelled as categorical dummies to capture non-linear premiums.

Heterogeneity: Urban vs. Rural

Does the market tightness matter?

- **Urban (Red):** Steep gradient. High willingness to pay for efficiency.
- **Rural (Green):** Flat gradient. Efficiency has little impact on price.



Section 5

Discussion & Conclusion

Limitations

- **Cross-Sectional Data:**
 - We observe a snapshot in time.
 - We cannot observe rent changes for the same apartment (no causal interpretation).
- **Omitted Variable Bias:**
 - Unobserved quality (e.g., “luxury” status) might correlate with energy efficiency.
- **Offer vs. Transaction:**
 - We analyze advertised rents, not final signed contracts.

Key Findings & Policy Implications

- **The Green Premium is Real:**
 - Energy efficiency is capitalized into rental prices.
- **Market Heterogeneity:**
 - The effect is largely driven by Urban markets.
- **Policy Implications:**
 - **Urban:** Market forces naturally reward decarbonization.
 - **Rural:** The market signal is weak. Government subsidies are critical to incentivize renovations.

Thank You!

- Questions?

Section 6

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